

ICD-10 Automated Codification

Task Analysis & Key Challenges

Fundamentals of NLP
UAB · 2025-26

THE TASK

"shock septico"



N189

Map short Spanish medical literals to ICD-10 codes
(both diagnosis & procedure codes)

DATA AT A GLANCE

13.7K

Training
literals

6.7K

Test
literals

4,059

Unique
codes

180K

Reference
ICD pairs

KEY CHALLENGES

47%

Extreme long-tail distribution

47% of codes appear only once; 76% appear ≤ 3 times

25%

Noisy & multilingual input

Abbreviations (HTA, RAO), typos, mixed Spanish/Catalan

19%

Ambiguous literals

19% of literals map to multiple different codes

2 sys

Mixed coding systems

64% diagnosis + 36% procedure codes with different structures, plus ICD-9 legacy codes

45% of test literals have exact matches in training · Average literal length: 2.2 words · Evaluation via Kaggle leaderboard

Proposed Solution: Multi-Strategy Cascade Pipeline

A cascading approach that routes each literal to the best-suited method



STEP 0 · Preprocessing

Normalize abbreviations · Fix typos · Detect language (ES/CA) · Split diagnosis vs. procedure codes

STEP 1



Exact & Fuzzy Match

~50%

estimated coverage

Direct lookup for literals seen in training.
Levenshtein distance for near-matches (typos, variants).

Highest confidence



STEP 2



Semantic Retrieval

~30%

estimated coverage

TF-IDF with char n-grams against 180K official ICD descriptions. BM25 ranking for best match.

Handles rare codes



STEP 3



ML Classification

~20%

estimated coverage

TF-IDF + SVM/Logistic Regression for codes with ≥5 training examples. Hierarchy-aware features.

Best for common codes



Next steps:

ICD hierarchy exploitation (parent-child inheritance for rare codes) · Ensemble voting across strategies · Deep learning (multilingual transformers) for follow-up 2